All-Weather Smart Vehicle Gate Recognition System

YOLOv8 · EfficientNet · Raspberry Pi 5 · ANPR · VMMR

Project Description

All-Weather Smart Vehicle Gate Recognition System is a project that focuses on providing access to vehicles based on their number plate and car model through authorization via a database. The UI stores the records and provides the admin with access to view logs and monitor live camera activity.

For implementation, the number plate is detected by training a YOLOv8 model followed by a number plate recognition model trained on an annotated dataset. The authorization is also influenced by vehicle make and model recognition, for which we created a large dataset and trained our model using EfficientNet.

In this project, the number plate is detected and recognized along with the vehicle make and model in all weather conditions. We are currently implementing it on a Raspberry Pi 5 (8GB), which will be connected to the camera installed at the gate.

The camera will capture the vehicle image, the Raspberry Pi procesess the image using the trained models, checks the vehicle details with the database, and then decides whether the vehicle should be allowed access or not.

Literature Review

1. Vehicle and License Plate Recognition with Novel Dataset for Toll Collection (2025) Uses cameras at toll points to automate recognition of vehicle types and license plates. Dataset: 10,000 images. YOLOv5/v8/v11 achieve ~99% accuracy; nano versions ~97–98%. Also deployed on Raspberry Pi.	Our Novelty We achieve 99.4% accuracy for license plate recognition. We classify by specific make and model (e.g., Honda Civic 2023) rather than broad categories. Our UI stores timestamps of authorized vehicles and provides live admin monitoring.
2. ANPR for Pakistani License Plates: A Deep Learning Approach Using CNN and YOLO (2025) Addresses fake plate detection challenges specific to Pakistani plates (non-standard sizes, fonts, colors). Used YOLOv3 on 200 images with 96.7% accuracy. Performed detection, segmentation, and recognition.	Our Novelty We achieved 99.4% on 600+ manually annotated images. We also perform VMMR trained on a large dataset. Our system includes a database, UI, and Raspberry Pi implementation. Their system detects fake plates, which our s currently does not.
3. Towards Efficient Vehicle Recognition: A Unified System for VMMR, ANPR, and Color Classification (2025) MobileNetV2 for VMMR, YOLO+EasyOCR for ANPR, HSV clustering for color. Dataset: 4,000+ images for YOLO, 364 Pakistani images for VMMR. Accuracies: ANPR 93.3%, VMMR 90.6%, color 71.8%.	Our Novelty Our YOLOv8 achieves 99.4% ANPR vs their 93.3%, avoiding EasyOCR's poor performance on Pakistani fonts. Our VMMR covers a far wider vehicle market. We include a UI, database, and edge device deployment.
4. An Efficient Method to Locate License Plate Under Diverse Environments (2025) Used Chinese CCPD dataset (290,000 images) with CNN, pooling, regional proposal network, and SGD	Our Novelty Their model is trained on Chinese plates and won't generalize to Pakistani plates. We surpass their 97.5% with 99.4% on harder Pakistani plates using YOLOv8. We

optimizer. Achieved 97.5% mAP, outperforming YOLOv3, Faster-RCNN, and RPNet on Chinese plates.	also add VMMR, a web dashboard, and edge device processing.
5. Number Plate Recognition Smart Parking Management System Using IoT (2025) Detects plates at entry/exit, calculates parking duration, generates billing. Uses Google Vision AI Cloud (98% accuracy), ESP32 microcontroller, IR sensors, and a web dashboard with database.	**Our Novelty** Google Vision AI is costly; we use a self-hosted solution at 99.4% accuracy. We use Raspberry Pi instead of ESP32 for its significantly higher processing power.
6. An Efficient Automated Vehicle License Plate Recognition System Under Image Processing (2023) MATLAB-based ALPR using grayscale conversion, noise removal, character segmentation with 36 classes. Dataset: 120 images. Detection 97.5%, recognition 96.67%, character recognition 94.17%.	**Our Novelty** We achieved 99.4% using character segmentation on a much larger dataset. Their system lacks VMMR and performs poorly under low light and rain due to the small training set.
7. Intelligent Vehicle Number Plate Recognition System Using YOLO for Enhanced Security in Smart Buildings (2024) Grayscale conversion and edge detection for plate detection. Used 37,000-image character/number dataset. Achieved 96% precision and 95% OCR accuracy. Fails at different angles and lighting conditions.	**Our Novelty** Our 600+ real plate images outperform their 37,000-image character-font dataset in real-world conditions. Their dataset doesn't train on real plates, causing failures at angles and distance. No VMMR, dashboard, or edge device.
8. Real-Time Car Model and Plate Detection System Using Deep Learning Architectures (2024) MobileNetV2+YOLO for VMMR; PaddleOCR+SVTR-tiny for ANPR. Tested at Firat University gate (Turkey). Detects only 30 car models from a 1,000-image dataset. Achieves 97.5% accuracy under fog/rain/low light.	**Our Novelty** Pakistani plates are harder than Turkish plates. Their VMMR covers only 30 models; ours is far broader. They outperform us in varied weather as they were trained on real-time weather data. No UI, database, or edge device.
9. Automatic Vehicle Number Plate Recognition System Using Tensor Flow Object Detection (2024) ANPR for Nigerian plates using TensorFlow+EasyOCR on a 203-image Kaggle dataset. Tested on 40 plates (38 correct). EasyOCR accuracy: 93.33%. Uses SQLite database for owner record lookup.	**Our Novelty** Small dataset and lower accuracy (93.33%). No VMMR, no UI, no edge device implementation.
10. An End-to-End Automatic Number Plate Recognition System Based on YOLOv12 and Tesseract OCR (2025) Turkish ANPR tested at Protel company gate on 1,200 entries. Dataset: 2,000 Kaggle + 500 custom images. YOLOv12: 96% plate detection, Tesseract OCR: 82% character accuracy. 3.2s/vehicle. PostgreSQL + TCP/IP gate control.	**Our Novelty** Tesseract OCR's 82% accuracy is very low and degrades further under fog/rain/low light. Pakistani plates are harder than Turkish. No VMMR, no UI, no edge device implementation.

ANPR Detection Results

The following images demonstrate successful license plate detection and recognition across various Pakistani vehicle types, lighting conditions, and plate styles.



Fig. 1 — Suzuki Mehran: MND464



Fig. 2 — Toyota Corolla: MN484



Fig. 3 — Mercedes E-Class: LEF743



Fig. 4 — Honda City: AHF743



Fig. 5 — Suzuki Swift: AEA591

System Authorization Output

The following screenshots demonstrate the complete end-to-end pipeline, including vehicle model classification, plate recognition, and authorization status.



Fig. 6 — Suzuki Mehran: RND-5050 UNAUTHORIZED



Fig. 7 — Suzuki WagonR 2015: APC-197 UNAUTHORIZED

YOLOv8 Validation Metrics

The heatmap below summarizes the key performance metrics of the trained YOLOv8 license plate detection model.

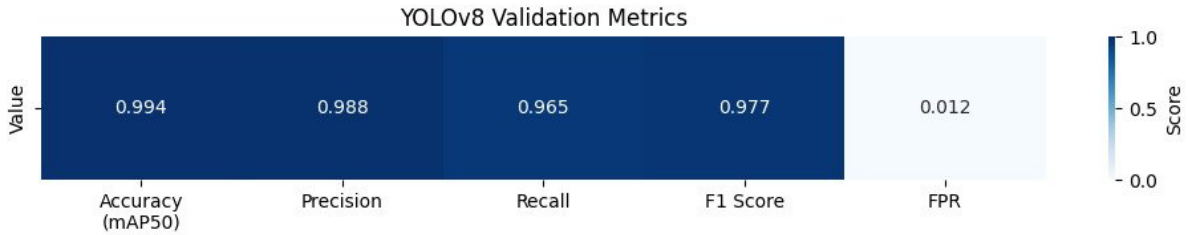


Fig. 8 — YOLOv8 Validation Metrics Heatmap (Accuracy 99.4% · Precision 98.8% · Recall 96.5% · F1 97.7% · FPR 1.2%)

Metric	Value	Interpretation	Benchmark
Accuracy (mAP50)	99.4%	State-of-the-art	Best in reviewed papers
Precision	98.8%	Very low false positives	Exceeds all compared systems
Recall	96.5%	Excellent detection rate	High coverage across plate types
F1 Score	97.7%	Strong balance	Robust under varied conditions
False Positive Rate	1.2%	Minimal incorrect detections	Near-zero gate errors